

An open source FORTRAN90 Code for aerodynamic design of wind turbine blades - V6 *

M. Korjahn, bewind GmbH, Rendsburg, Germany
A.P. Schaffarczyk, Kiel University of Applied Sciences, Kiel, Germany
D. Schlipf, Flensburg University of Applied Sciences, Germany
contact: Alois.Schaffarczyk@HAW-Kiel.de

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Abstract

A specific aerodynamic design methodology has been implemented into an open source FORTRAN90 code (*KSS-Blade*). It is available via github: <https://github.com/Schaffarczyk/KSS-Blade/>. Version 5 added the radial induced velocity for non-planar (deflected or prebent) rotors following Li et al. [16] in two variants, Madsen's engineering correction and a vortex-cylinder model, and removed a number of latent coding inconsistencies found by a systematic (LLM-assisted) static analysis. Version 6 corrects the non-planar formulation of V5 in four respects that were identified during the public discussion of the accompanying brief communication in *Wind Energy Science* [31]: the axial momentum balance of the tilted annulus ($\cos^2 \kappa$), the torque integration along the blade, the sign of the radial-velocity force for downwind deflection, and the cylinder strengths (annulus instead of blade induction). The accompanying axial term of the vortex-cylinder model, which compensates the radial term in the integral but not locally, is evaluated in post-processing. All V5 results of section 9 are superseded; the claim of a grid-dependent tip value is withdrawn. A new flag *NPform* in *Machine.in* allows to reproduce the V5 formulation.

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1 Introduction

Here we report on an independent FORTRAN 90/95 (F90 for short) implementation of a purely aerodynamic wind turbine blade design methodology. It is intended to be as detailed as possible although a fair amount of knowledge of wind turbine aerodynamics is necessary, see, for example, [35, 33, 32]. Some – but not all – needs from a more practical point of view (maximum chord and twist, etc.) have been taken into account. This is in contrast to [13] who tried to find out what maximum c_P may be reached without any restrictions by *practicability*.

Our basis, a lecture of Korjahn [15], presents a so-called two-point design. In a first attempt, however, we only describe an implementation of the first part: to arrive at a blade with as large a portion of the span at optimum $a(r)$ ¹ as possible. *Design point 2* consists of a certain safety against gusts which correspond to stalled parts on the blade. Therefore the AOA has to have some safety margin against stall.

Main objectives of our approach are:

- understand and formalize the approach and
- compare to other codes.

Further or additional sources from

1. Korjahn [15],
2. Bak [2],
3. Schmitz [33] and
4. Johansen et al. [13].

are recommended to complement understanding.

Important: The code is by no means fool-proof:

Bugs may still be present.

Input data and esp. the aerodynamic one has to be checked as far as possible before its use.

2 The Blade-Element Momentum (BEM) formulation

2.1 General

(Conservation of) momentum on independent annular rings [35, 26, 29] serves as the fluid mechanical foundation. Key elements of a complete BEM formulation are

¹Usually the term Betz-optimum is related to $a(r) = 1/3 = \text{const.}$ Very recently [12] it has been shown that including wake-rotation and tip-loss a somewhat different distribution results: Close to the hub $a \rightarrow 0.25$ and close to the tip $a \rightarrow 0.4$.

already discussed in [7] and further refined in classical texts by Wilson et al. [29] and de Vries [26], for example. Any applied aerodynamic wind turbine investigation uses it and it is still today regarded as the *work horse*. CFD can hardly claim to be more accurate although a large number of assumptions inherent to the BEM model have been overcome by this very different approach. The most recent analysis of the BEM-method is given in [10] with some focus on mathematical rigor and optimization.

2.2 Multiple Solutions of the BEM-Equation

It was already recognized by Wilson et al. [29] p 108 ff. that under some specific circumstances the BEM-equations may not produce unique solutions. Maniaci [5] and Ning [30] further elaborate on this subject. To give an example, Fig. 1 (for a special section of a special commercial blade) shows the local thrust coefficient cT_{loc} as function of a , the axial induction, defined by $u_1 := (1 - a) \cdot U_{\infty}$, $0 \leq a \leq 0.5$. Momentum balance gives

$$c_T = 4a \cdot (1 - a), \quad (1)$$

whereas the counter-force from lift is its proper projection into the direction of the wind. Note that this is an approximate description only, as a' (circumferential induction) is neglected as well as the tip correction (see section 2.3) and a correction of c_T for larger values of a ($a > a_c$), see section 2.4.

Our investigation for a 20 MW blade of 133 m length confirms this. It was recognized that the traditional fixed-point iteration then did not converge but simply jumped between two different values.

As a consequence we improved our implementation and search algorithms as follows:

- Firstly, scan the whole a -region $0 \leq a < 1$ in steps of 0.01. If there are solutions found by a binary search algorithm,
- start using standard fixed-point iteration until a maximum number of iterations is reached,
- switch to Newton(-Raphson) for further iterations if no convergence is reached, and finally
- use the binary search algorithm *rtbis* from [27].

Fig. 2 shows an example. Therefore the recommendation of [29], p. 31

As a guiding principle, it is suggested that the solution to be chosen in the case of multiple solutions is the one that maintains continuity of angle of attack along the blade span.

can be confirmed.

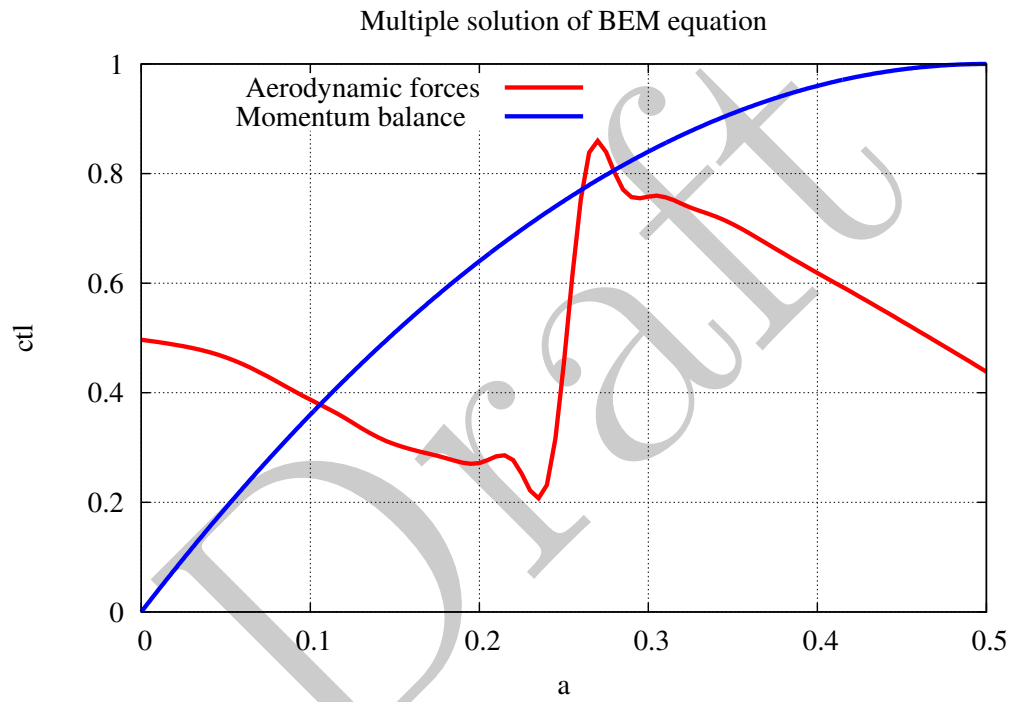


Figure 1: Example of distributions of multiple solutions from balancing aerodynamic (lift) forces and local (axial) momentum.

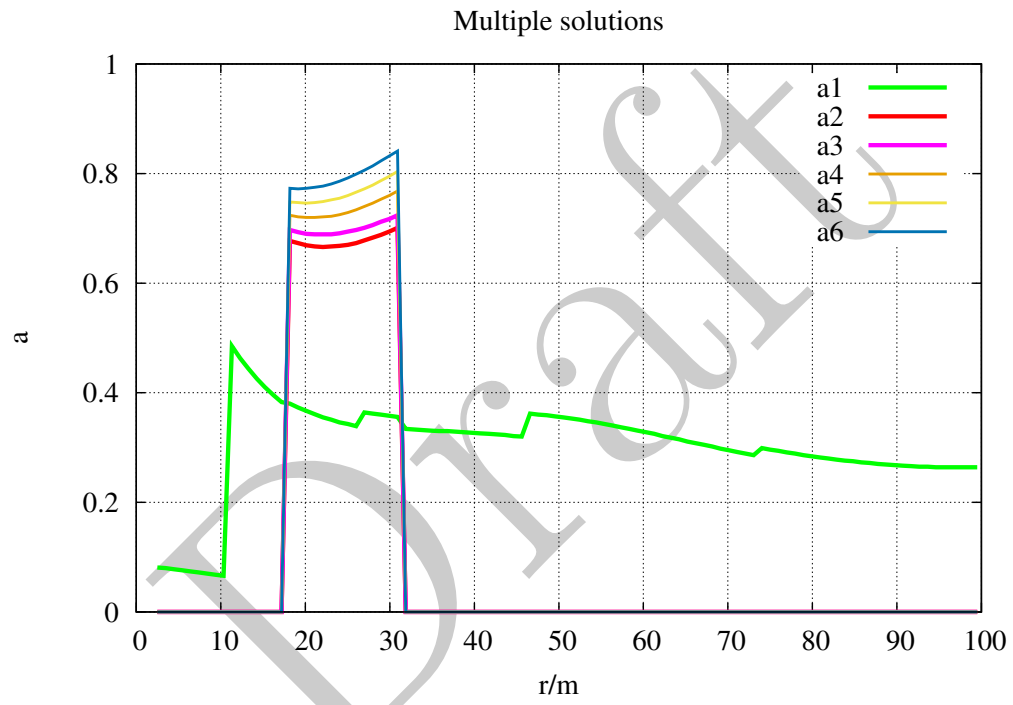


Figure 2: Example of how a binary search algorithm finds multiple solution along a whole blade (CIG10MW)

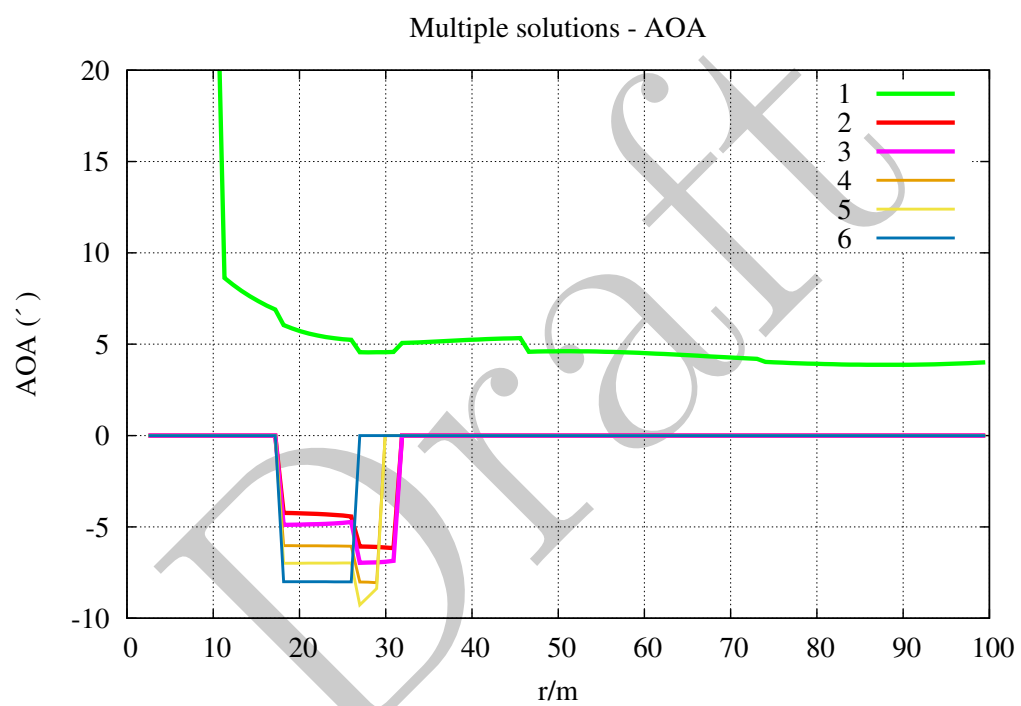


Figure 3: Variation of Angle of Attack (CIG10MW)

2.3 Tip-correction

Momentum theory is strictly valid only for an infinite number of blades, $B \rightarrow \infty$. As in most designs B is small ($B < 4$) this has to be taken into account. Indeed, as was shown very early, *tip losses* are as large as profile drag and must be accurately embedded into the code. Traditionally, the simplest (and oldest) version uses a factor F , which reduces circulation to zero when $r \rightarrow r_{tip}$:

$$F = \frac{2}{\pi} \arccos \left(\exp \left[-\frac{(B/2)(1 - r/R_{tip})}{r/R_{tip} \sin(\varphi)} \right] \right) . \quad (2)$$

Here, φ is the all-important flow angle defined by

$$\tan(\varphi) = \frac{1 - a}{\lambda(1 + a')} , \quad (3)$$

which refers to $\lambda = \omega \cdot r / U_{\infty}$ the local Tip-Speed-Ratio (TSR). $\omega = RPM \cdot \pi / 30$ is the angular velocity.

2.4 Glauert Correction $cT(a)$ for $a > a_c$

It is further well known:

- Momentum theory is valid for $0 \leq a \leq 0.5$ only, and
- for some a_c Eq. 1 becomes inaccurate.

To some extent this can be cured by *empirical engineering* extensions which are listed, for example in [10], Table 2.1. In principle at some $a_c = 0.33...0.4$ a more or less linear curve is attached to Eq. (1) so that $c_t(a = 1)$ approaches (somehow) 2 for $a \rightarrow 1$.

2.5 Sign convention

Sign issues esp. for the pitch are notorious. Following our basic interpretation

$$\alpha \text{ (AOA)} = \varphi \text{ (flow angle)} - \vartheta \text{ (twist)} , \quad (4)$$

we define **pitch** as being added to twist.

3 Code Description

3.1 A short note on FORTRAN

We will not defend our choice of the *grandfather of all (higher) programming languages* but simply use it. We have tried to write the code as plainly as possible to make understanding and changes (**i.e. maintainability**) as simple as possible. A minimum set of algorithmic features is used. In particular, no tricks from computer science have been used. This code was (and is) developed using *gfortran*, version 4.8.5 from 2015, see section 3.7. A more detailed description is given in [23].

3.2 Numerical Approach

Everybody knows from their own experience the NEWTON(-Raphson) method for finding zeros of functions ($f(x_0) = 0$) or SIMPSON's rule for 1D-Integration. The ultimate manifestation of numerical algorithms may be found in the famous (series of) text book(s): *NUMERICAL RECIPES in Fortran 77* [27]. From here we have copied

1. SPLINE and
2. SPLINT.

Unfortunately, some overshooting at sharp corners might still be possible. *Bezier* polynomials were coded independently from its definition see [24].

3.3 Main program

KSS.f contains the main code. It is merely a collection of routine calls for set-up and definition of the course of the investigation:

1. (dynamic) memory allocation,
2. Read *Machine.in*,
3. Read in spline thickness distribution from *ThickDis.in*,
4. Analyze a given set of profiles, L2D-max.
5. If design-mode is chosen:
6. An initial BEM calculation is performed and
7. chord and twist might be (Bezier-) smoothed or
8. twist may be changed to improve c_P , see section 6.1 .

3.4 Memory management

mem.f defines all GLOBAL data. Each program unit using it must start with a
use mem
statement.

3.5 Subroutines

1. BEM: BEM iteration as described in [32],
2. ANALYSIS: Analyses a blade described in *BlaDes.in* using routine *BEM*,
3. DESIGN: Determines chord and twist from given profiles and Betz [32] optimum criteria. See section 6.1 for more information about optimization.,

4. AIRFOILS: searches given α, c_L, c_D data for maximum of c_L/c_D , $c_{L,max}$ and slope at zero lift.
5. TipShapeEL: defines an elliptic (vertical tangent) tip shape geometry,
6. TipShapePara: defines an parabolic tip shape geometry determined by smooth slope at the beginning of the tip region and zero chord at tip.
7. BEZIER: smoothing of a given data set according to the definition of Bezier polynomials,
8. smoothCHORD: smooths chord with subroutine BEZIER,
9. smoothTWIST: smooths twist with subroutine BEZIER,
10. improveTWIST: (1) tries to get locally as close as possible to $1/3$ by adaption of twist, see Eq. 9. (2) J.N. Sørensen's approach, (3) a *constant circulation* approach

3.6 Functions

1. FNOVERK: (*Real Function N Over K*) calculates Binomial coefficients (BC) needed in BEZIER, (special numerical care has to be taken because of the strong growth of BCs),
2. FPR: Prandtl's tip loss F, see [32],
3. FBu: Burton's tip loss, see [28, 4],
4. chDES: optimum chord distribution from Betz (or Schmitz, or Glauert) [35],
5. chBe: Betz optimum chord, $a' \equiv 0$,
6. chSc: Schmitz optimum chord, a' is included,
7. TWIST: twist (in deg) from Betz, $a = 1/3$ and $a' = 0$, so only local TSR enters,
8. AHANSEN: gives Glauert's correction as stated in [9],
9. ANREL: gives Glauert's correction as stated in [19].

$$c_T(a) = \frac{8}{9} + \left(4F - \frac{40}{9}\right)a + \left(\frac{50}{9} - 4F\right) \cdot a^2. \quad (5)$$

10. thickpoly: cubic interpolation of thickness
11. twistbend: simple table for inclusion of twist-bend
12. apot: (1) Betz, (2) J.N. Sørensen, (3) constant circulation (very approximate)

3.7 Compilation

3.7.1 LINUX

For compilation we prefer to use (in a SUSE-LINUX environment): [23] with mandatory options (see **compDesCode.cmd**):

-fno-automatic helps to resolve notorious issues with local variables in SUBROUTINES and FUNCTIONS

-fbounds-check to avoid not allowed memory access

-O3 high optimization

A typical job with 100 sections and 100×100 wind-speed and pitch variation takes a few minutes on our simple *Intel(R) Core(TM) i3-4160 CPU @ 3.60GHz* CPU.

3.7.2 Windows

MinGW (= minimal gcc for Windows) was used to compile a Windows executable file, which is also available.

3.8 Modularization

Apart from mem and KSS the following files

- mem.f, declares global variables,
- Sub.f, a set of used subroutines and functions,
- SubNum.f, numerical functions: SPLINE and SPLINT from [27],

help to put some structure into the code.

3.9 Work Flow

All input has to be provided as ASCII files, and all output is written in the same format. Post-processing then has to be performed by other tools. We prefer GNU PLOT [25, 11] see section 4.2.

4 Output Files and Tools for Visualization

4.1 Description of Output Files

- *Bem.out*: sectional output from BEM, see table 5,
- *des.out*: sectional output from design: rsec, twist, chord, thick, c_L^{des} and AOA_{des} ,
- *ProProp.out*: from routine AIRFOILS: design properties of given airfoil data.

4.2 Description of GNUPLOT scripts

- *a.gpl*: plots $a(r)$. Data from *Bem.out*
- *chord.gpl*: plots $c(r)$. Data from *Bem.out*, *BlaDes.in* and *des.out*
- *cL-aoa.gpl* and *cL-cD.gpl*: plots polars. Data from **.aer*
- *cL-des.gpl* plots $c_L^{des}(r)$. Data from *PrOpt/Bem.out*
- *twist.gpl*: plots $\vartheta(r)$. Data from *Bem.out*, *BlaDes.in* and *des.out*
- and some more

5 Mode: Analysis

5.1 Input data

The code is able to operate in two modes:

1. Design mode, set bit *DesMode* to .T. in file *Machine.in*,
2. if not, then the code will operate to analyze a given blade.

Then, in addition, other files:

- *BlaDes.in* (r , twist, chord, attached profile name),
- *ProThick.in* which maps airfoil names to their (relative) thickness,
Important: This file has to have a header line
- *ThickDis.in* which gives with few entries r/R_{tip} and corresponding t/c ,
- **.aer*: files with airfoil data $AOA(\alpha)$, c_L , c_D , c_M (not used).

are needed. Most of the input should be more or less self-explanatory. Lines with 3 numbers are: first value, last value and number of values (wind speed and pitch).

Examples are given below.

Fig. (5) and Fig. (4) give an impression of what kind of thickness distributions have been and will be used.

Recently, [34] investigated a 15 MW reference blade in more detail. For reducing the mass from 71 t to 63 t a considerable change of thickness (see Fig. 6) was necessary. This underlines again the importance of this file.

Main output file is *Bem.out*

```

DesMode      .F.
BladeNo      3.0
dens         1.225
Rhub         0.3
Rtip         63.0
windrage     11.4 11.4 1
rpm          12.1
Pitchr       0.0 0.0 1
Nsec         70
. . . . .

```

Table 1: Part of a sample job description file *Machine.in* based on NREL’s 5MW reference wind turbine

```

r      Twist    chord    Profile Name
0.0000 15.5946  5.2000  CYL1
2.4000 15.5911  5.2083  CYL2
4.8000 15.4288  5.2357  FF50
7.2000 14.9891  5.2889  FF50
9.6000 14.3263  5.3606  FF50
12.0000 13.4947  5.4433  FF50
15.0000 12.3001  5.5510  FF50
18.0000 11.0323  5.6501  FF50
20.2841 10.0834  5.7110  FF36
. . . . .
118.7179 -1.3914  1.7239  FF21
118.9744 -1.3628  1.6717  FF21
119.2308 -1.3336  1.5813  FF21
119.4872 -1.3038  1.4157  FF21
119.7436 -1.2734  1.1027  FF21
120.0000 -1.2424  0.5000  FF21

```

Table 2: Sample blade description file *BlaDes.in* for NREL/IEAwind 15MW reference wind turbine

NAME	rel thickness
CYL1	1.0
CYL2	0.95
FF50	0.5
FF36	0.36
FF33	0.33
FF30	0.3
FF27	0.27
FF24	0.24
FF21	0.21

Table 3: Sample thickness assignment file *ProThick.in* for NREL/IEAwind 15MW reference wind turbine

0.0000	1.0
0.0200	0.98
0.1500	0.500
0.2452	0.360
0.3288	0.330
0.4392	0.301
0.5377	0.270
0.6382	0.241
0.7717	0.211
1.0000	0.211

Table 4: Sample Thickness distribution file *ThickDis.in* for NREL/IEAwind 15MW reference wind turbine

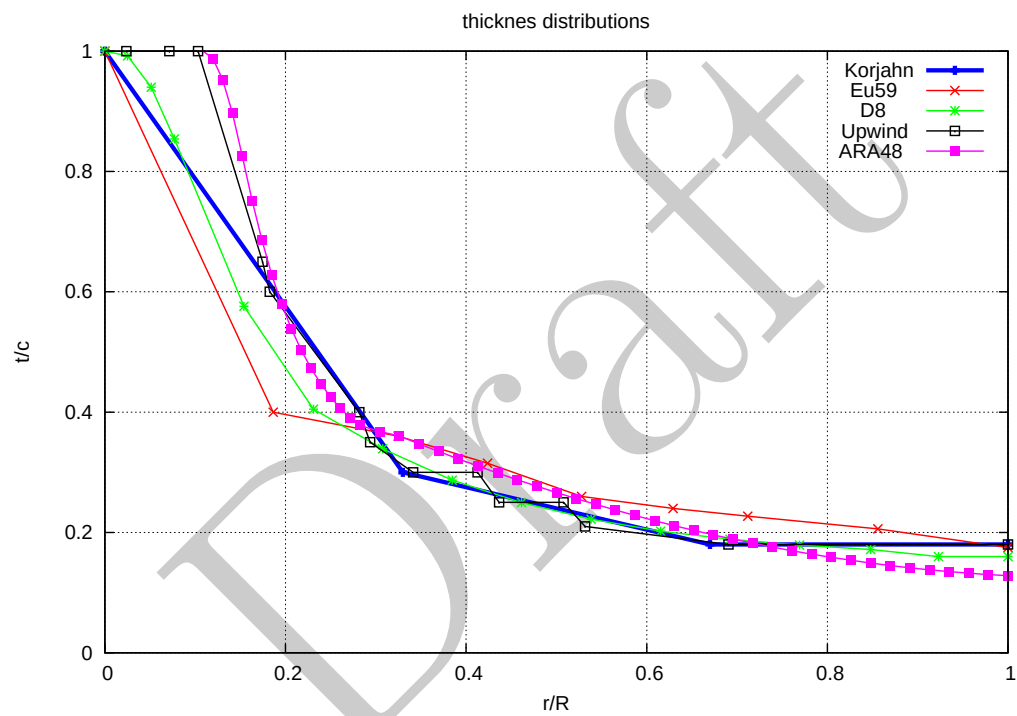


Figure 4: Sample thickness distributions

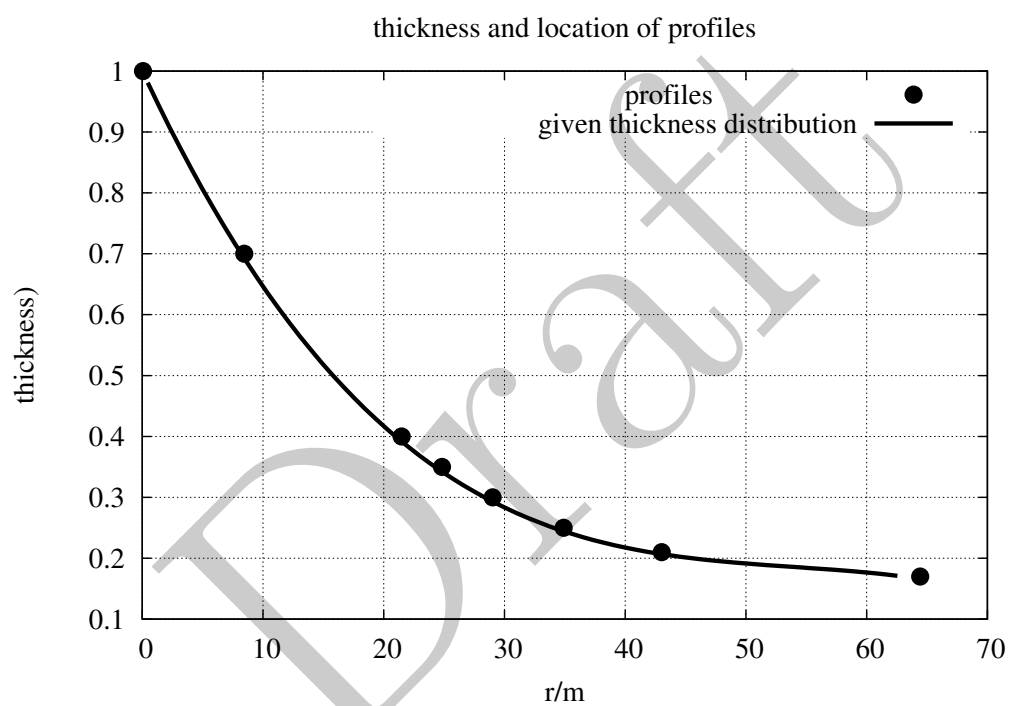


Figure 5: Refined thickness distribution from NREL 5 MW

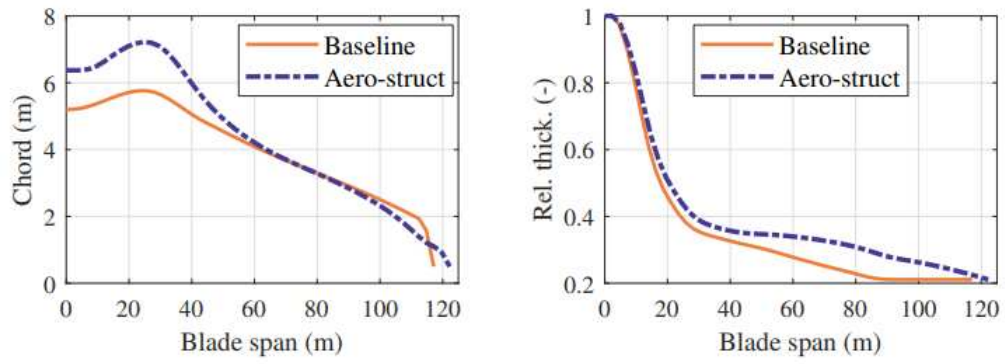


Figure 4: Chord and relative thickness distributions.

Figure 6: Thickness distribution change to decrease mass for a 15 MW reference blade [34]

r	a	aP	F	w	chord	twist	phi	aoa	cL	cD	L2D	cNo	cTa	dT	dQ	iter	err	th	Prof
0.598	0.334	-0.334	1.0	6.004	5.202	15.596	87.4	71.8	0.000	0.481	0.000	0.481	-0.022	0.20	-0.01	3	0.0E+00	0.99	CYL2
1.794	0.134	-0.134	1.0	7.861	5.205	15.598	82.3	66.7	0.000	0.444	0.000	0.440	-0.059	0.31	-0.04	3	0.0E+00	0.98	CYL2
2.990	0.076	-0.076	1.0	8.519	5.213	15.573	77.3	61.7	0.000	0.388	0.000	0.379	-0.085	0.31	-0.07	3	-0.2E-06	0.97	CYL2
4.186	0.045	-0.045	1.0	9.009	5.226	15.495	72.5	57.0	0.000	0.303	0.000	0.289	-0.091	0.27	-0.08	3	-0.2E-06	0.95	CYL2
5.382	0.040	-0.020	1.0	9.353	5.246	15.347	67.5	52.2	0.060	0.303	0.197	0.303	-0.061	0.31	-0.06	4	-0.1E-04	0.92	FF50
6.578	0.038	-0.002	1.0	9.730	5.273	15.128	62.8	47.7	0.144	0.297	0.484	0.330	-0.008	0.36	-0.01	4	-0.3E-04	0.89	FF50
7.774	0.040	0.011	1.0	10.149	5.305	14.848	58.4	43.5	0.245	0.283	0.864	0.370	0.060	0.44	0.07	5	-0.9E-07	0.85	FF50
8.970	0.043	0.021	1.0	10.608	5.340	14.519	54.2	39.7	0.356	0.263	1.354	0.421	0.135	0.56	0.18	6	0.0E+00	0.81	FF50
10.16	0.049	0.028	1.0	11.106	5.379	14.144	50.4	36.3	0.469	0.238	1.966	0.482	0.210	0.70	0.31	6	0.0E+00	0.76	FF50

Table 5: Sample output file *Bem.out* for NREL/IEA wind 15MW reference wind turbine

6 Mode: Design

If the design flag is switched on, the lower part of *Machine.in* is important and will be used. As the Betz and/or Schmitz chord distributions usually give non-zero chord at all sections, a tip length must be given within which the chord goes down to zero. Two shapes can be chosen:

- Parabolic "P" and
- Elliptic "E".

The latter gives a vertical tangent at the tip and the former is a simple quadratic parabola defined by two points (begin and end (=rtip)) and slope of $c(r)$ at the begin of the tip region.

Note: *ImpChord* flag is not used at the moment. Improvement of chord (to proceed to more monotonic-like behavior, see [15] sheet 11) is achieved by smoothing with *Bezier* polynomials.

In addition to Korjahn's $a = 1/3$ approach *DesSchema* = 1, we use *DesSchema* = (2,3) for other Schemas (with very low differences), as described in section 6.1.

6.1 Aerodynamic Optimization

6.1.1 Betz' and other criteria

We will not give a complete list or discuss in much detail the various interpretations of what an optimization might be and how this may be reached. Only the following three are implemented:

- Betz: $a(r) = 1/3 = \text{const.}$,
- Sørensen's (2022), modified *Glauert's* approach[12], see Fig. 7 and
- Joukowski's: $\Gamma(r) = \text{const.}$

(1) Betz: no further explanation needed.

(2) Sørensen [12] argued that close to the tip $a \rightarrow 2/5$, see Fig. 7.

(3) The last one is more subtle; to first order (in a) it is equivalent to Betz', but to 2nd order:

$$a(1 - a) = \frac{\lambda B \Gamma (1 + a')}{4\pi}. \quad (6)$$

To complete, a model for a' has to be added, for example

$$a' = \frac{1 - 3a}{4a - 1} \text{ (Glauert) or} \quad (7)$$

$$a' = \frac{2}{9} \frac{1}{\lambda_r^2} \text{ (Simple approximation) .} \quad (8)$$

DesSchema = 1 refers to Betz' aerodynamic optimized blade $a = 1/3$ for all spans.

.				
10.00000	-0.88209	0.01386	-0.04397	
-8.00000	-0.62981	0.01075	-0.05756	
-6.00000	-0.37670	0.00882	-0.06747	
-4.00000	-0.12177	0.00702	-0.07680	
-2.00000	0.12810	0.00663	-0.08283	
-1.00000	0.25192	0.00664	-0.08534	
0.00000	0.37535	0.00670	-0.08777	
1.00000	0.49828	0.00681	-0.09011	
2.00000	0.62052	0.00698	-0.09234	
3.00000	0.74200	0.00720	-0.09447	
4.00000	0.86238	0.00751	-0.09646	
5.00000	0.98114	0.00796	-0.09828	
6.00000	1.09662	0.00872	-0.09977	
7.00000	1.20904	0.00968	-0.10095	
8.00000	1.31680	0.01097	-0.10163	
9.00000	1.42209	0.01227	-0.10207	
.				

Table 6: Sample aerodynamic data *FF21.aer* for NREL/IEAwind 15MW reference wind turbine

.	
Tiplen	3.0
twmax	20.
chmax	5.5
ImpChord	.F.
ImpTwist	.F. 1
ImpThick	.F.
TwistB	.F.
DesSchema	2

Table 7: Parameters describing a design case

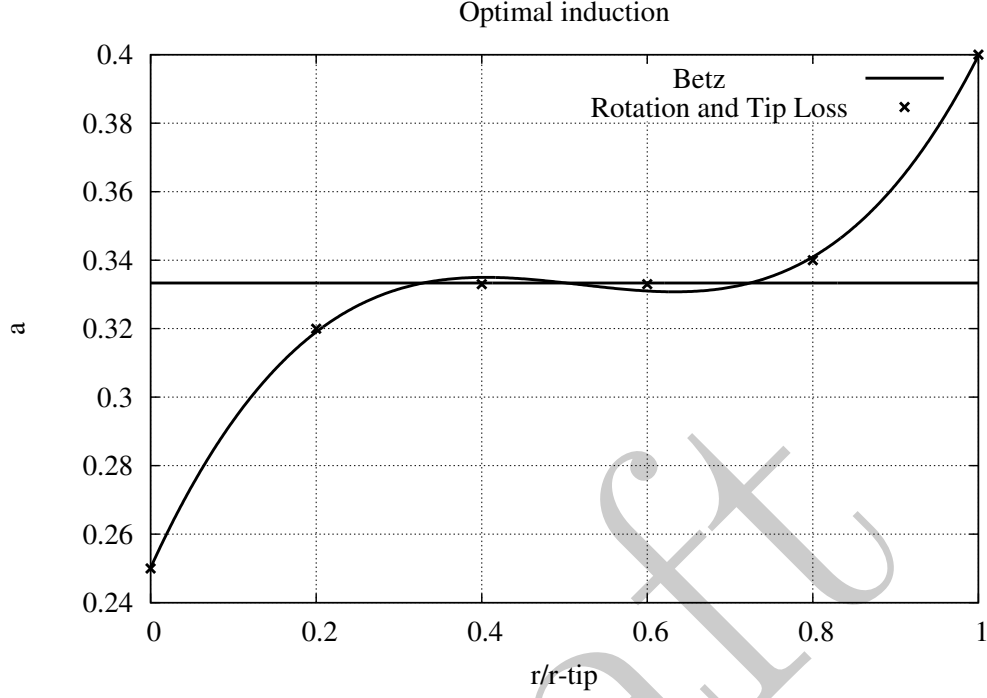


Figure 7: Betz and Sørensen

DesSchema = 2 refers to Sørensen's different $a(r)$ distribution, [12]. Fig. 7. See [13] for further discussion about aerodynamic optimization and the textbooks in the list of references, esp. [33].

DesSchema = 3 is implemented but seems not to give any significant changes. In any case we will only change twist for optimization.

6.1.2 Change of Twist

Our approach to relate twist and a runs via the flow angle and reads as: If $da := a_{opt}(r) - a_{loc} \neq 0$ then locally change twist by

$$d\varphi = -\frac{1}{\lambda_r(1 + a')}da . \quad (9)$$

6.1.3 Change of Thickness - Arrangement of Profiles

As stated in [15] (sheet 19, first figure) a further means to improve performance is to reduce thickness in the inner part. This is provided in our code by setting the *ImpThick*-bit and providing a changed thickness distribution *ThickDis.new*.

At the very end of *Machine.in* a flag *TwistB* (binary) can be set. It is a very simple way to take *pre-twist* into account. If set to .T., a table of additional twist deformation (along the blade) as a function of wind speed has to be given.

7 Sample cases

7.1 Case 1: The NREL 5MW Baseline blade

As this turbine [14] was also used by Korjahn [15] as a sample case, we first report on output from KSS as well.

Table 8: Rated rotor power of NREL 5MW baseline from various sources. HAWC data from Mr. Hinrichs, aerodyn engineering GmbH; bladed data from Mr. Maniok, DNV. If all values are equally weighted, $P = 5356 \pm 89$ kW results.

Source	Value/kW	Deviation (%)
TP 38060	5267	0.0
KSS	5248	-0.4
FAST	5293	0.5
bladed	5322	1.0
HAWC2	5323	1.1
q-blade	5364	1.8
Xturb	5418	2.9
WZX	5478	4.0
wt-perf	5494	4.3

Except for the (outdated) *wt-perf* (and some other codes) accuracy is within a 1 % range. The difference of KSS to TP38060 is only 0.4 %. This may serve as a hint that this BEM implementation is not totally wrong.

To have a glimpse of the *design capability*, Fig. 8 shows $a(r)$ after 5 loops of twist optimization. cP improves from 0.4682 to 0.4685. We agree with [15], sheet 16, that

For radius < 15 m optimum induction is not possible.

In addition, Fig. 9 shows some angles for further assessment.

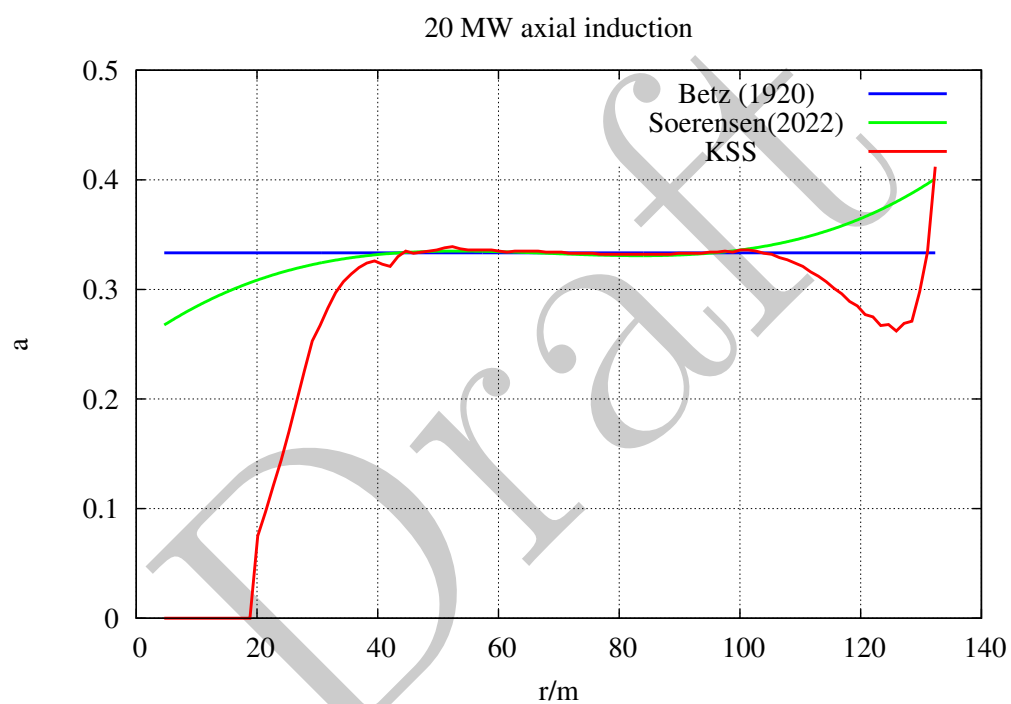


Figure 8: KSS Optimisation after 5 twist loops: $a(r)$

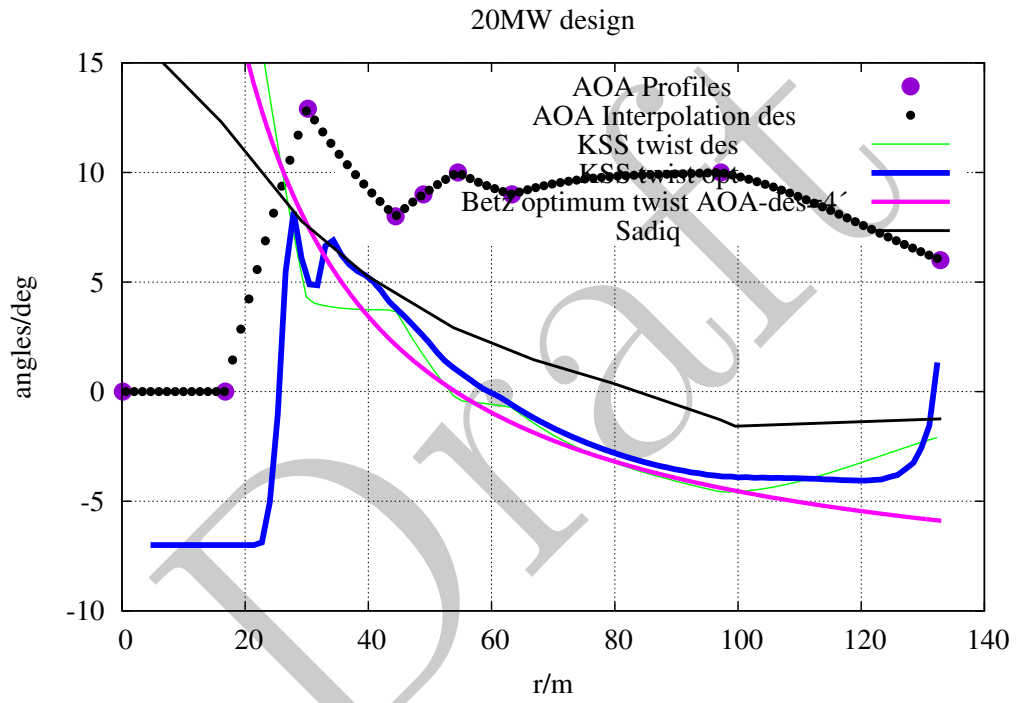


Figure 9: KSS Optimisation (Own 20MW, $r_{tip} = 133m$) after 5 twist loops: some angles

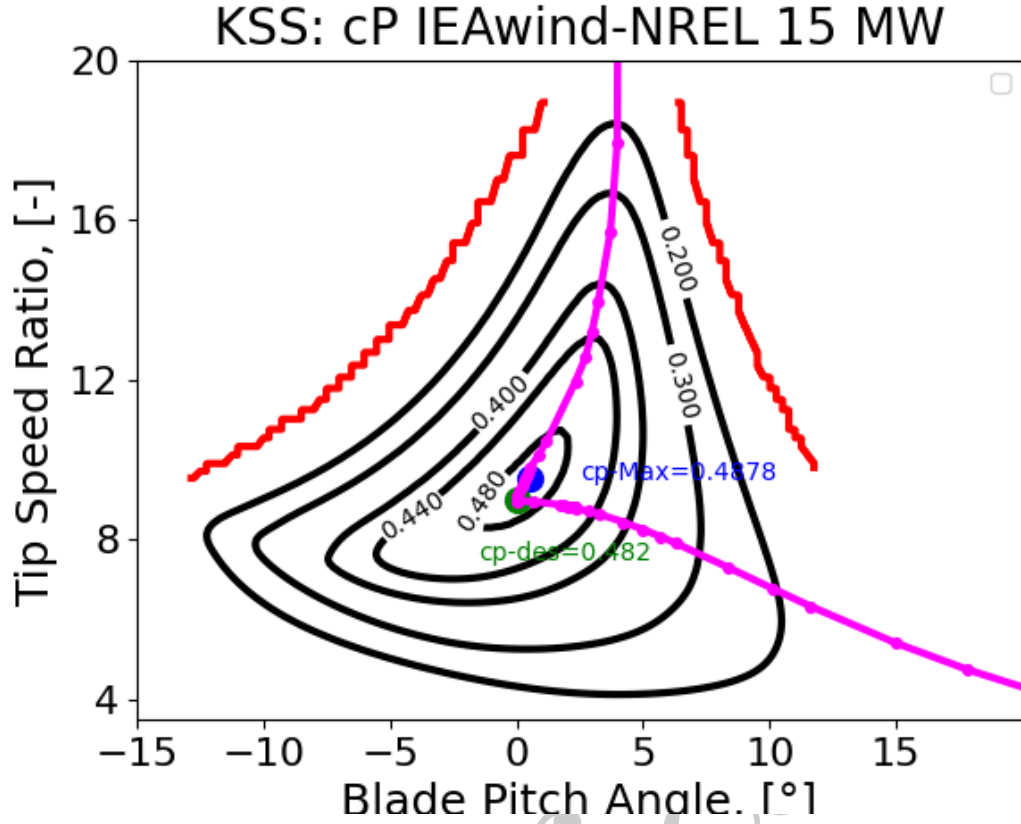


Figure 10: c_P map from KSS, including control path (magenta) and an extended pitch variation $-15^\circ < \vartheta < 20^\circ$.

7.2 Case 2: A 15MW WT reference blade

The blade described in [6] was also analyzed with KSS, see Figs. 10 and 11. Some differences are visible, but agreement at rated conditions, see table 9 (comparing c_P and c_T), is again within the 1% range.

Table 9: Comparison for NREL/IEAwind 15 MW reference wind turbine

Source	c_P	c_T
TP 76698	0.489	0.804
KSS	0.484	0.768

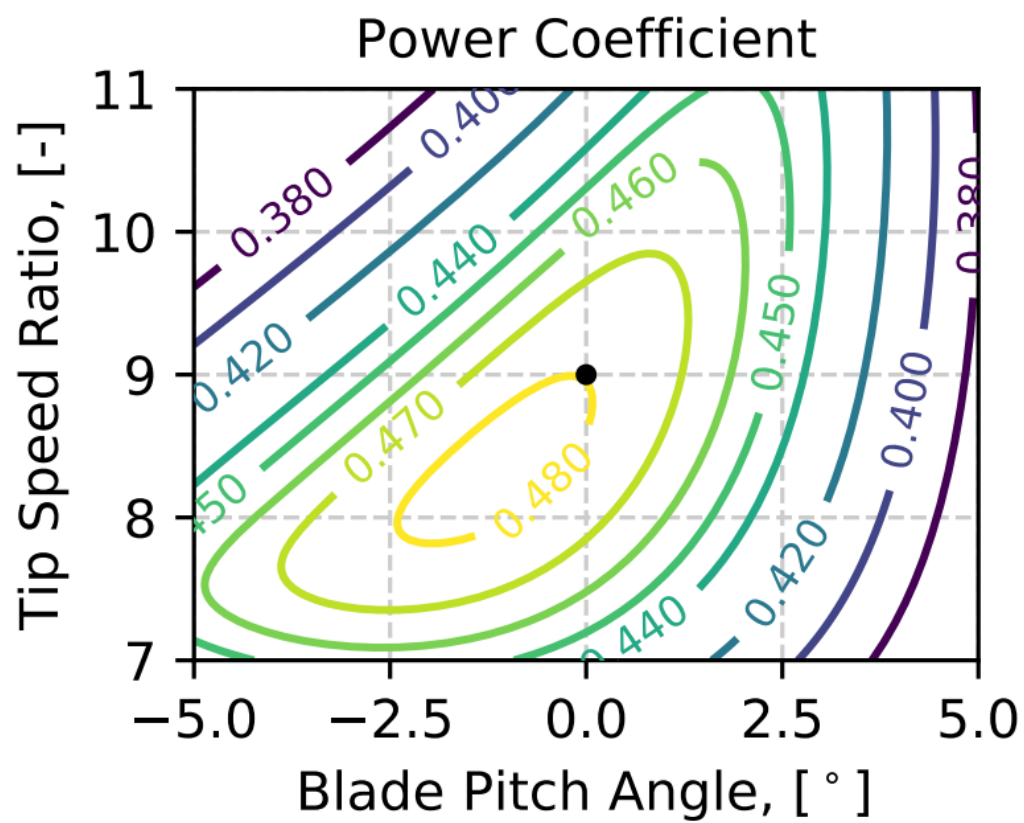


Figure 11: cP map from NREL [6]

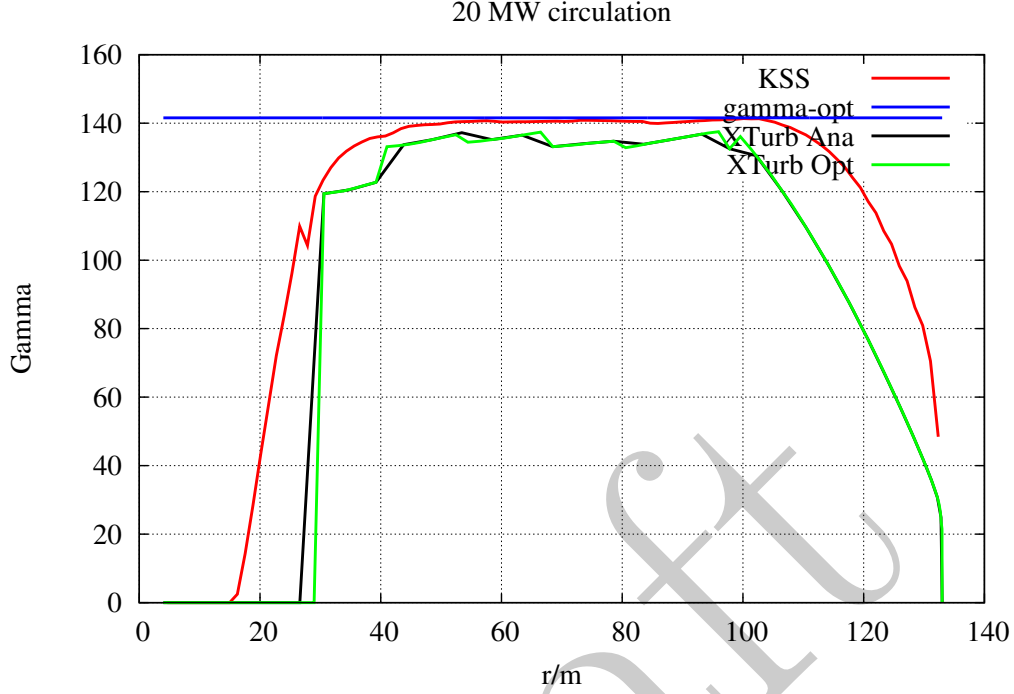


Figure 12: Comparison of circulation for an own 20 MW blade

7.3 Case 3: A 20 MW blade

[36] investigated the aerodynamic design of a 20MW blade in more detail. A maximum $cP = 0.4826$ was reached by Betz-mode of optimization after 6 loops and an initial design with $cP = 0.4821$. This is less than 1%. It seems that the restrictions are so strong that noticeable improvement is difficult.

8 Comparison with *Xturb*

Xturb by Sven Schmitz [33] offers some tools for optimization as well. We compared our 20 MW design [36], see Fig. 12. Unfortunately, XTurb showed an unexpected and non-explainable decrease close to tip ($r/R_{tip} > 0.77$) which makes a reasonable comparison impossible. Nevertheless, the initial design can be improved from $cP_{xT,A} = 0.4299$ to $cP_{xT,D} = 0.4347$ which is about 1.1% higher.

9 V5/V6: Radial induction for non-planar rotors

9.1 Motivation and history

Classical BEM treats the rotor as planar and retains only the axial and circumferential induced velocities. For blades with substantial out-of-plane (oop) deflection or prebend — i.e. all recent multi-MW designs — the radial induced velocity u_r , which accompanies wake expansion, is no longer orthogonal to the local blade section and produces an additional in-plane (driving or braking) force proportional to $\tan \kappa$, where κ is the local dihedral angle of the blade axis with respect to the rotor plane. Li et al. [16] give a computationally efficient framework for this effect; see also Madsen [20, 21, 22] and the vortex cylinder model of Branlard and Gaunaa [3]. Blade sweep is not considered.

Two members of this model hierarchy were implemented in V5 within the scope of the MSc thesis project of B. Masare and reported in a brief communication submitted to *Wind Energy Science* [31]. The public discussion of that paper (community comment by J. G. Leishman, two anonymous referee reports, Sept./Oct. 2026) — in particular a source-level analysis of the V5 code by referee 1 — showed that the non-planar formulation of V5 was inconsistent in four respects and that the radial-velocity term alone is not a meaningful estimate of the power change of a non-planar rotor. V6 corrects these points. This section describes the V6 formulation; the differences to V5 are collected in section 10.2 and in Table 11. The sample case throughout is a *modified* NREL 5 MW rotor, see section 9.7.

9.2 Input: new flags in *Machine.in* and *BlaDes.in*

Three lines were appended to *Machine.in*:

Radial	.T.	main switch (.F. = planar BEM)
RadMode	2	0 = deflected geometry only, no u_r term
		1 = Madsen correction (Eqs. 6+7 of Li et al.)
		2 = vortex cylinder (Sec. 3
		of Li et al.)
NPform	1	1 = consistent non-planar formulation (V6)
		0 = formulation of V5 (reproduces the discussion paper)

Table 10: New *Machine.in* flags in V5/V6. Missing lines default to *Radial* = .F., *RadMode* = 1 and *NPform* = 1, so old input files run unchanged.

The oop deflection $z_{oop}(r)$ (in m) is read from a fifth column appended to *BlaDes.in*. It is **positive downwind**, at unchanged projected radial stations, so that the projected tip radius R is kept. The local dihedral angle $\kappa = \arctan(dz_{oop}/dr)$ is evaluated by a central difference at the section midpoints (V5: backward difference, lagging by $dr/2$). Following [16], $\kappa > 0$ denotes **upwind** dihedral; a downwind deflection therefore gives $\kappa < 0$. Coning is the special case of constant κ .

9.3 Non-planar BEM (NPform = 1)

With the projected radius r , the blade length element $ds = dr / \cos \kappa$ and the section relative velocity W with $W \sin \varphi = U_0(1 - a) \cos \kappa$, the axial momentum balance of an annulus, $B f_n \cos \kappa ds = 4\pi r \rho U_0^2 a(1 - a) F dr$, and the torque read

$$\frac{a}{1 - a} = \frac{B c C_n \cos^2 \kappa}{8\pi F r \sin^2 \varphi}, \quad dQ = B f_t r ds, \quad (10)$$

where f_n and f_t are the section-normal and tangential forces per unit blade length and C_n the normal-force coefficient. The $\cos^2 \kappa$ factor is applied in all three branches of the induction iteration (fixed-point, Glauert/Hansen correction, Newton), the Glauert correction for $a > 0.4$ as before. V5 used the section kinematics with $\cos \kappa$ but the *planar* momentum balance and integrated the torque over dr ; with NPform = 0 this behaviour is reproduced.

The induction of Eq. (10) is the blade induction a_B . The annulus induction a_∞ follows from the same thrust coefficient without tip loss,

$$4a_\infty(1 - a_\infty) = 4F a_B(1 - a_B), \quad (11)$$

(Glauert branch as for a_B) and is written to *VC.out*.

9.4 Radial-velocity term

The radial induced velocity is added once per section *after* the induction iteration has converged (one-way coupling in the nomenclature of [16]; no feedback into the a -iteration). The additional in-plane force per blade follows from Kutta–Joukowski with the bound circulation $\Gamma = \frac{1}{2} W c c_L$,

$$dF_{z,inc} = \rho \Gamma u_r \tan(\kappa) dr, \quad \kappa > 0 \text{ upwind.} \quad (12)$$

An outward u_r thus *drives* a blade with upwind dihedral and *brakes* a blade deflected downwind. In V5 the downwind deflection was inserted with the wrong sign, so that all power changes attributed to u_r had the wrong sign.

9.5 Model 1: Madsen correction (RadMode = 1)

The area-averaged thrust coefficient up to radius r ,

$$c_{T,av}(r) = \frac{\int_0^r c_T(r^*) 2\pi r^* dr^*}{\pi r^2}, \quad (13)$$

drives the radial induced velocity (exact logarithmic form of Eq. (6) in [16]),

$$u_r(x) = \frac{U_0}{2.24} \cdot \frac{c_{T,av}}{4\pi} \ln \left[\frac{0.04^2 + (x + 1)^2}{0.04^2 + (x - 1)^2} \right], \quad x = r/R. \quad (14)$$

The Madsen expression is defined inside the rotor disc only and has no axial counterpart.

9.6 Model 2: Vortex cylinder (RadMode = 2)

The converged *annulus* inductions $a_{\infty,i}$ of the BEM sweep define $N_{sec}+1$ semi-infinite right vortex cylinders of strength

$$\gamma_t^{(k)} = 2U_0 (a_{\infty,k} - a_{\infty,k-1}), \quad a_{\infty,0} = a_{\infty,N+1} = 0, \quad (15)$$

located at the annulus boundaries R_k and starting at the axial position of the *deformed* blade (non-planar rotor, Sec. 3.3 of [16]). V5 used the blade induction a_B instead, which does not vanish at the tip (see section 9.9). The radial velocity induced by one cylinder at evaluation point (r, x) , x being the axial distance to the start of the cylinder, is [3]

$$u_r(r, x) = -\frac{\gamma_t}{2\pi} \sqrt{\frac{R_k}{r}} \left[\frac{2-k^2}{k} K(k^2) - \frac{2}{k} E(k^2) \right], \quad k^2 = \frac{4rR_k}{(R_k+r)^2 + x^2}, \quad (16)$$

with complete elliptic integrals K and E (polynomial approximation [1], 17.3.34–37, error $< 2 \cdot 10^{-8}$; the modulus is limited to $k^2 \leq 1 - 10^{-6}$). u_r at a blade section is the superposition of all cylinders; the force increment again follows from Eq. (12). Sectional details are written to the output file *VC.out* ($r, a_B, a_{\infty}, \kappa, u_r^{VC}, dF_{z,inc}$).

9.6.1 Axial term (post-processing)

The same cylinders induce an axial velocity u_x (closed form in [3]). For the planar rotor their sum reproduces a_{∞} exactly; for the non-planar rotor the axial induction at the blade changes by $\Delta a(r) = -(u_x^{np} - u_x^{pl})/U_0$. At fixed circulation the power change of the *complete* VC model with respect to the BEM solution of the deflected rotor is [17]

$$\Delta P = B\rho\Omega \int_0^R \Gamma r [-U_0 \Delta a + u_r \tan \kappa] dr, \quad (17)$$

i.e. the radial term of Eq. (12) plus an axial term. **The axial term is not part of the FORTRAN code.** It is evaluated by the Python scripts *vcfull.py* and *decomp.py* (double-precision elliptic integrals, verified against direct Biot–Savart integration of vortex rings) from the KSS output of the deflected rotor. This one-way evaluation at fixed circulation is not the coupled iteration of [16].

9.7 Sample case: modified NREL 5 MW rotor

The rotor is a modified NREL 5 MW rotor [14]: the blade is described from a root radius of 0.3 m (hub radius of the reference turbine: 1.5 m), three stations at 60.0, 62.32 and 63.0 m with the chord tapered to 0.01 m are added at the tip, and the two cylindrical root sections (*CYL1.aer*, *CYL2.aer*) use $c_D = 1.0$ (reference: 0.50 and 0.35). Note that these two polar files were missing in the *NREL-5MW* folder of the GitHub repository (V5); with the IEA-15MW cylinder polars ($c_D = 0.5/0.3$) V5 gives 692.2 kN / 4751.8 kW instead of the values of Table 11, which explains the reproducibility offset reported by referee 1.

The oop deflection is $z(\xi) = d f(\xi)$, $\xi = (r - r_0)/(R - r_0)$, $r_0 = 2.87$ m. Three shapes are used: linear, $f = \xi$ (the reference case with $d = 6$ m downwind, equivalent to a straight blade coned by $\kappa = -5.7^\circ$), a cantilever bending line, $f = \xi^2(3 - \xi)/2$, downwind, and an outboard-concentrated prebend, $f = \xi^3$, upwind. Operating point of the reference case: $v = 11$ m/s, 12.1 rpm, pitch 0° , $N_{sec} = 70$.

9.8 Results: reference case

Table 11 gives the V5 and the V6 results side by side.

Table 11: Modified NREL 5 MW rotor, 11 m/s, 12.1 rpm, 6 m linear downwind tip deflection, $N_{sec} = 70$. Thrust / kN and power / kW for the formulation of V5 ($NPform = 0$) and V6 ($NPform = 1$). The last row (complete VC model) is evaluated in post-processing at the fixed circulation of the *RadMode 0* solution.

	V5 (NPform 0)		V6 (NPform 1)	
	Thrust	Power	Thrust	Power
planar (<i>Radial .F.</i>)	699.77	4718.40	699.77	4718.40
deflected, no u_r (RadMode 0)	695.12	4650.46	697.92	4716.52
+ radial term, Madsen (RadMode 1)	695.12	4726.29	697.92	4640.08
+ radial term, VC (RadMode 2)	695.12	4728.71	697.92	4620.54
+ radial and axial term, VC	–	–	697.92	4715.85

The corresponding c_P values (planar / deflected / Madsen / VC / complete VC) are 0.4642 / 0.4575 / 0.4649 / 0.4652 / – for V5 and 0.4642 / 0.4640 / 0.4565 / 0.4545 / 0.4639 for V6. With the consistent formulation the deflected geometry alone changes the power by -1.9 kW only (thrust -0.3 %); the -67.9 kW of V5 were an artefact of the inconsistent momentum balance and torque integration. The radial term alone *reduces* the power by 76.4 kW (Madsen) and 96.0 kW (VC), i.e. by 1.6 and 2.0 %, the two models differing by 26 %. In the complete VC model this loss is compensated by the axial term of Eq. (17), $+95.3$ kW, because the blade sections of a rotor deflected downwind lie upstream of the starting points of the outboard vortex cylinders and experience a smaller axial induction. The net change is -0.7 kW (-0.01 %), in agreement with the independent calculation of referee 1 ($+92.3$ and -91.8 kW with strengths from the lift-only thrust coefficient). **The radial term on its own is therefore not a meaningful estimate of the power change of a non-planar rotor**, and the statement of V5 ($\Delta c_P / c_P \approx +0.2$ %) is wrong in sign, magnitude and interpretation.

Fig. 13 compares the local distributions. The two u_r models cross at $r \approx 0.68R$; on the outer quarter of the blade the VC value exceeds the Madsen value by a factor of 1.3 to 1.9, reaching 4.3 m/s at the outermost section against 2.3 m/s (6.5 m/s with strengths from a_B as in V5). At mid-span the VC result changes sign, following the non-monotonic $a_\infty(r)$ of the BEM solution ($\gamma_t \propto \Delta a$), which the smooth $c_{T,av}$ -driven Madsen fit cannot represent: u_r is directed inward between $r = 6$ and 31 m with a

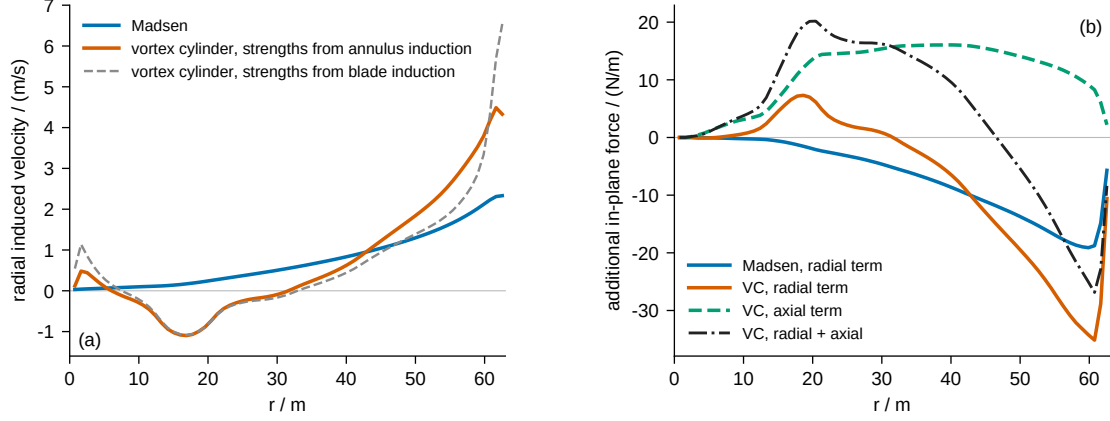


Figure 13: Modified NREL 5 MW, 11 m/s, 6 m linear downwind tip deflection, $N_{sec} = 70$ (V6). (a) Radial induced velocity $u_r(r)$ from the Madsen model and from the VC model with strengths from a_∞ and from a_B (V5). (b) Additional in-plane force per blade and unit radius: radial term, Eq. (12), for both models, axial term of Eq. (17) and their sum for the VC model (positive: driving).

minimum of -1.1 m/s. The sign change persists when $a(r)$ is smoothed; its origin is the axial induction increasing outward over the lightly loaded root sections, and it is not a coding error.

Fig. 13b shows the corresponding in-plane forces. The radial terms of the two models differ by a factor of 1.8 in the peak sectional force (-35 against -19 N/m at $r \approx 61$ m) and by a factor of 1.9 in the moment contribution of the outer 5 % of the blade (-15.3 against -7.9 kNm, converged for $N_{sec} \geq 140$). The axial term of the VC model is distributed quite differently, with its maximum at mid-span. Radial and axial terms therefore cancel in the integral but not locally: the complete VC model shifts in-plane loading from the outer 10 % of the blade (-30 kW) to the inner part ($+29$ kW), with a peak change of -27 N/m, about 6 % of the local aerodynamic in-plane force.

9.9 Tip behaviour and cylinder strengths

V5 reported a logarithmic growth of the VC value at the outermost section with the number of sections, $u_r \simeq 0.78 \ln N_{sec} + 3.0$ (m/s), and called it an N_{sec} -dependent artefact of the discrete cylinder sum. This growth is reproduced in Fig. 14a (squares) and is a consequence of forming the cylinder strengths from a_B : a_B does not vanish at the tip (0.23 for the $N_{sec} = 70$ solution), so the tip cylinder keeps a constant strength $|\gamma_t|/2\pi = 0.81$ m/s while the evaluation point approaches it as $1/N_{sec}$, where $K(k^2) \simeq \frac{1}{2} \ln[16/(1-k^2)]$. With the strengths formed from a_∞ as in [16], the tip-loss factor lets a_∞ and hence the tip cylinder strength vanish at the tip ($|\gamma_t|/2\pi$ decreases from 0.38 to 0.02 m/s for $N_{sec} = 35 \dots 4480$, roughly as $N_{sec}^{-1/2}$) and the outermost value converges: 4.12, 4.33, 4.32 and 4.27 m/s for $N_{sec} = 35, 70, 140$ and 280 in re-converged BEM runs, and 4.32 to 4.35 m/s for $N_{sec} = 70 \dots 4480$ with

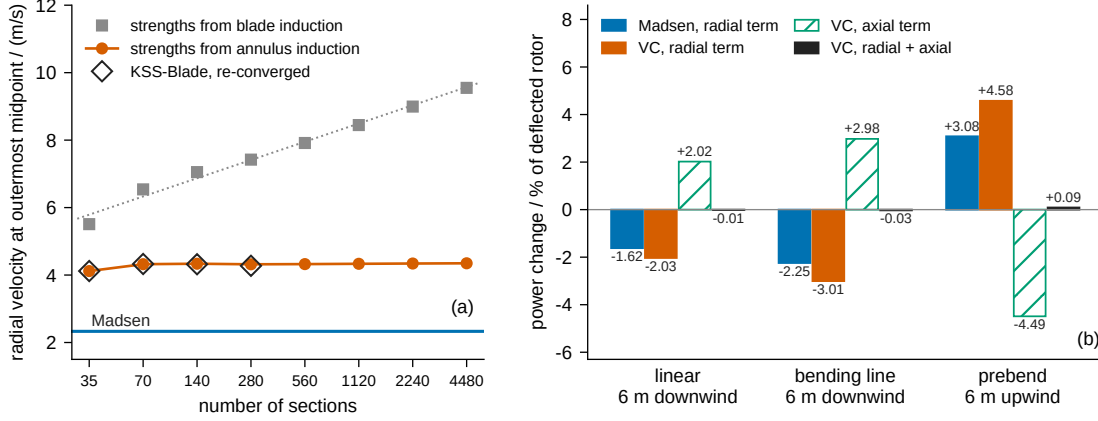


Figure 14: (a) VC radial induced velocity at the outermost section midpoint versus N_{sec} with cylinder strengths from a_B (squares, V5; dotted: logarithmic fit) and from a_∞ (circles; fixed $a_B(r)$ and $\varphi(r)$ of the $N_{sec} = 70$ solution), re-converged BEM runs (diamonds) and Madsen value (line). (b) Power change relative to the deflected rotor at 11 m/s for three deflection shapes with 6 m tip deflection.

$a_B(r)$ and $\varphi(r)$ fixed. **The VC formulation of [16] is thus not grid-dependent at the tip; the statement to the contrary in V5 is withdrawn.** The limit $k^2 \leq 1 - 10^{-6}$ in the FORTRAN routine, which in V5 turned the spurious growth into a plateau, now lowers the tip value by less than 0.02 m/s for $N_{sec} \leq 280$.

9.10 Dependence on loading and deflection shape

Both models were run for five operating points along the steady-state schedule of the turbine (7, 9, 11 m/s at 8.0, 10.3, 12.1 rpm and zero pitch, $c_T \approx 0.77$; 13 and 15 m/s at 12.1 rpm, 6.6° and 10.45° pitch, $c_T = 0.39$ and 0.24), tip deflections of 3, 6 and 9 m and the three shapes of section 9.7, 45 cases in all (Fig. 14b, scripts *study4.py*). The radial term scales linearly with the tip deflection and changes the power by -0.2 to -4.7 % for the downwind shapes and by up to $+7.1$ % for the 9 m upwind prebend (VC model). Its magnitude is larger in the VC model than in the Madsen model throughout: by 25–27 % (linear), 32–36 % (bending line) and 46–53 % (prebend) at high loading and by 12–23 % above rated wind speed. In the complete VC model the axial term compensates the radial term in every case; the net change lies between -0.05 and $+0.14$ % of the power. The agreement of the two models in the integral that V5 observed for the linear shape is thus a cancellation that holds for constant κ only.

9.11 Consistency checks suggested in the discussion

Limacher and Wood [18] proved for an actuator disc that $\int_0^\infty (a^2 - u_r^2/U_0^2) r dr = 0$ in the rotor plane (see also [38, 37]). For the planar rotor the VC field satisfies this relation to four significant digits, for a uniformly loaded disc as well as for the

present loading at 7, 11 and 15 m/s (*lw_check.py*); about half of $\int u_r^2 r dr$ lies outside the rotor disc. The Madsen expression is not defined outside the disc; inside, its integral is only 34–36 % (7 and 11 m/s) and 65 % (15 m/s) of the VC value. A further remark concerned the role of drag [8]: here the cylinder strengths follow from the thrust coefficient including drag, and Γ in Eq. (12) is the bound circulation that carries the lift. Strengths from the lift part alone change u_r by less than 0.07 m/s for $r > 0.5R$ and the net power change from -0.7 to $+0.7$ kW; reducing Γ by the factor $1 - (c_D/c_L) \cot \varphi$ (4–8 % for $r > 0.5R$) reduces both terms of Eq. (17) by 6–7 %.

9.12 Limitations and open points

- **One-way coupling:** cylinder strengths are formed from the *converged* BEM inductions and the axial term is evaluated at fixed circulation; the coupled formulation of [16], Sec. 3.2, is the appropriate basis and remains to be implemented.
- **Axial term only in post-processing:** the FORTRAN code (RadMode 1, 2) evaluates the radial term only. Used alone it changes the power by several per cent in the wrong direction; the Python scripts must be run for the complete VC model. The Madsen correction has no axial counterpart; combined with a planar momentum balance it gives misleading power and load changes.
- **Sign changes:** in the mid-span region u_r^{VC} may become negative (inward). This is physical and not a coding error.
- Only u_r enters the FORTRAN code via Eq. (12); the axial VC induction does *not* replace the momentum balance of the BEM.

10 Code revisions V5 and V6

10.1 V5: consistency fixes

In preparation of V5 the complete source was checked statically (LLM-assisted review, 2026 July). The following latent inconsistencies — most of them uninitialized variables that stayed unnoticed due to *-fno-automatic* zero-initialization — were removed:

1. *Sub1.f*: u_r , $c_{L,0}$, $c_{D,0}$ were used *undefined* in the first pass of the former in-loop radial correction; block replaced by the post-convergence correction of section 9.
2. *rtbis*: interpolation weight was never set (new module variable).
3. *ctsec*: a_c and x (tip loss argument) undefined; airfoil interpolation used the same polar twice with interchanged weights.

4. Duplicated output and limiter blocks removed.
5. *Airfoils*: undefined output unit number; index of the zero-lift search could stay undefined for cylinder profiles.
6. *fnoverk*: the fixed-form line `if (2*k.gt.n) then k=n-k` was parsed as an assignment to an implicit variable `thenk` (!) — i.e. the binomial symmetry shortcut was silently never active; corrected using a local copy of the dummy argument.

None of these fixes changed the reference results of Table 8 beyond the last printed digit, but they remove undefined behaviour.

10.2 V6: consistent non-planar formulation

V6 (October 2026) corrects four points of the non-planar formulation of V5 that were identified by referee 1 of the WES discussion [31] by a source-level analysis, plus one further inaccuracy found during the revision. All are active for $NPform = 1$ (default); $NPform = 0$ reproduces V5.

1. *Sub1.f*: axial momentum balance of the annulus with $\cos^2 \kappa$, Eq. (10), in the fixed-point iteration, the Glauert/Hansen branch and the Newton branch (V5: planar balance).
2. *Sub1.f*: torque integrated over the blade length $ds = dr / \cos \kappa$ (V5: over dr).
3. *Sub1.f*: sign of the radial-velocity force. The oop deflection in *BlaDes.in* is positive downwind; an outward u_r then *reduces* the driving force, $dF_{z,inc} = -\rho \Gamma u_r \tan \kappa dr$ in code variables (V5: wrong sign).
4. *Sub1.f*, *Sub3.f*: vortex cylinder strengths from the annulus induction a_∞ , Eq. (11), instead of the blade induction a_B ; a_∞ is written to *VC.out*.
5. *Sub1.f*: κ from a central difference about the section midpoint (V5: backward difference, lagging by $dr/2$).

mem.f and *KSS.f* were extended by the flag $NPform$ and by $RadMode = 0$. The planar results ($Radial = .F.$) are unchanged. Differs against V5 (*V5-to-V6-Sub1.diff*, *V5-to-V6-Sub3.diff*) and the Python scripts of all studies (*study3.py* reference case and grid study, *study4.py* parametric study, *study5.py* local loads and drag, *lw_check.py*, *vcfull.py*, *decomp.py*, *figs2.py*) are part of the repository. Build:

```
gfortran -std=legacy -fno-automatic -O1 -o kss \
    mem.f KSS.f Sub1.f Sub2.f Sub3.f SubNum.f
```

11 Summary of investigated blades

Table 12 shows a summary of investigated blades so far:

Table 12: Summary of investigated blades

Name	Rated Power	Rotor Diameter/m
SkyWind NG	0.6 - 1 kW	1.5
Baltic Thunder	~ 1 - 5 kW	1
Braak	~ 10 kW	101
SWT	1 kW	1.5
SWT(WINDFLOH)	14 kW	10.3
NASA-AMES	15 kW	11
E30	150 kW	30
GROWIAN	1000 kW	100
NecMicon 80	2 MW	80
MM92	2 MW	92
NREL-Baseline	5 MW	126
DTU	10 MW	178
CIG10MW	10 MW	200
		220
NREL	15 MW	239
	17 MW	248
KUAS20	20 MW	284
22-RWT	22 MW	266

12 Summary, Discussion and Outlook

Based on the lecture notes from [15] an F90 blade analysis and design tool has been implemented. It was applied to and tested on a wide range of WT blades, see table 12. Agreement with other sources/codes within a 1% range or better seems possible.

Findings:

- Choice of thickness distribution and polar data has probably the strongest influence on performance data for a wind turbine blade.
- Improvement (of c_P) by the proposed 1-point algorithm seems to be within a 1%-range only.
- Non-planar rotors (V6): the Madsen correction and the vortex-cylinder model give radial induced velocities of the same order, but they differ by a factor of about two on the outer 5 % of the blade and in sign at mid-span, and their power-relevant radial terms differ by 12–53 % depending on loading and deflection shape. The radial term must not be used alone: it changes the power of the NREL 5 MW rotor by -4.7 to $+7.1$ % for tip deflections up to 9 m, whereas the complete VC model (radial plus axial term) predicts changes below 0.14 %, with a redistribution of in-plane loading along the span (see section 9). The V5 statement of a $+0.2$ % gain and of a grid-dependent tip value is withdrawn.

13 Disclaimer

As mentioned earlier, a set of **meaningful** input data has to be provided. There is almost no **consistency** check of input data and this implies that the user has to have some knowledge to avoid crashing of the code and/or meaningless output. Again and again we emphasize:

Theorem 1. *You can't calculate what you haven't understood.*

Theorem 2. *Only start the calculation, if you know the answer.*

14 List of Abbreviations

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Table 13: list

	meaning
CIG	Chinese, India, Germany
DNV	Det Norske Veritas
DTU	(The) Danish Technical University
HAWC	Horizontal Axis Wind turbine simulation Code
KUAS	Kiel University of Applied Sciences
NREL	National Renewable Energy Laboratory (USA)
SWT	small wind turbine

source code, the inconsistencies of the non-planar formulation and the missing axial term; the V6 results rest on this analysis.

Draft

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